

# ECON 1550 International Finance

## The II-XX Model

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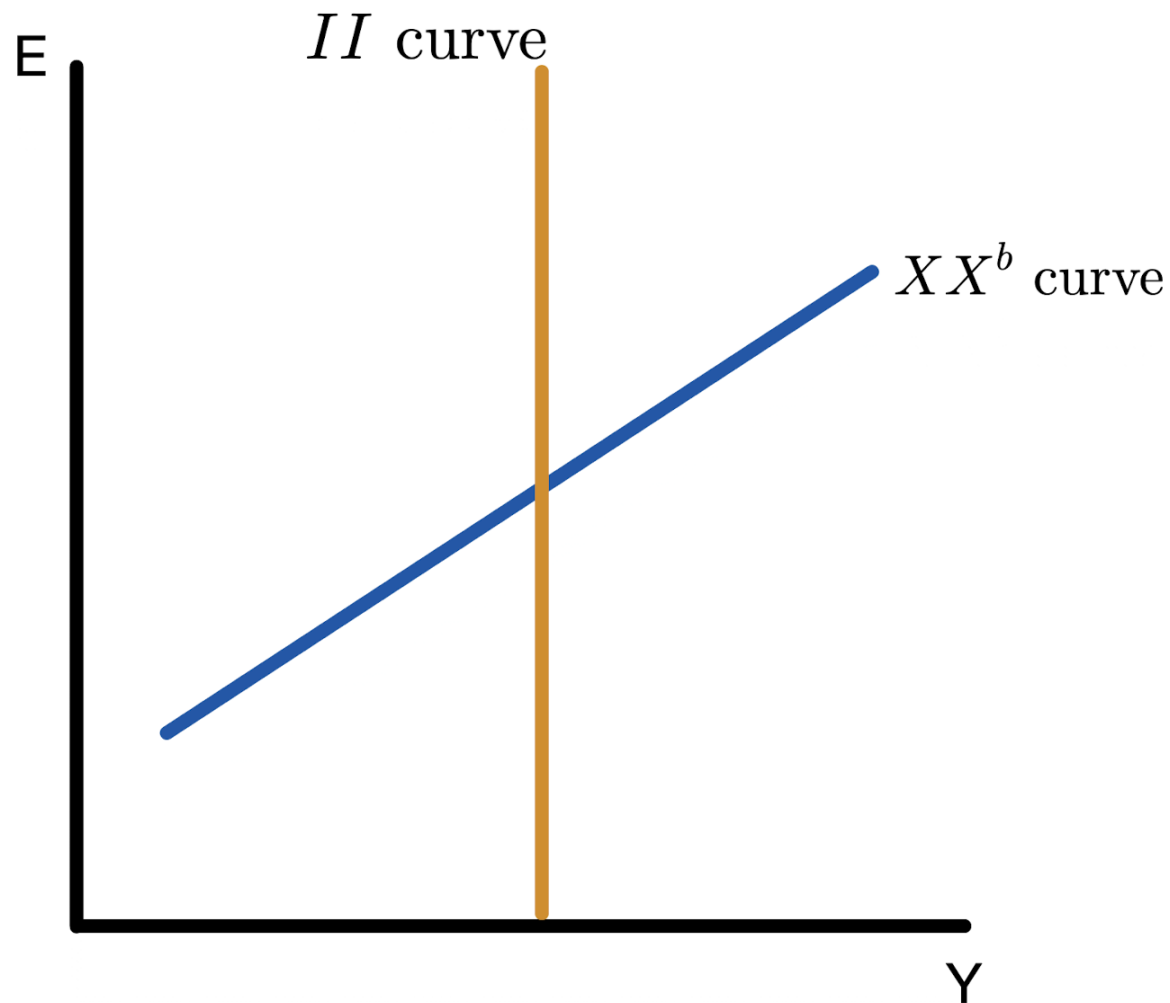
- Start with minimal assumptions
  - There are numbers  $Y^f$  and  $XX^b$  such the economy tends to  $Y = Y^f$  and  $CA = XX^b$  in the long run
  - $Y = Y^f$  and  $CA = XX^b$  are desirable economically or politically
  - Monetary and fiscal policy cannot influence  $Y^f$  and  $XX^b$ , but can affect how long it takes for  $Y$  and  $CA$  to get to  $Y^f$  and  $XX^b$

# II-XX Model

- Do not assume
  - Equilibria in any markets
  - Mechanisms for how  $Y$  and  $CA$  approach  $Y^f$  and  $XX^b$

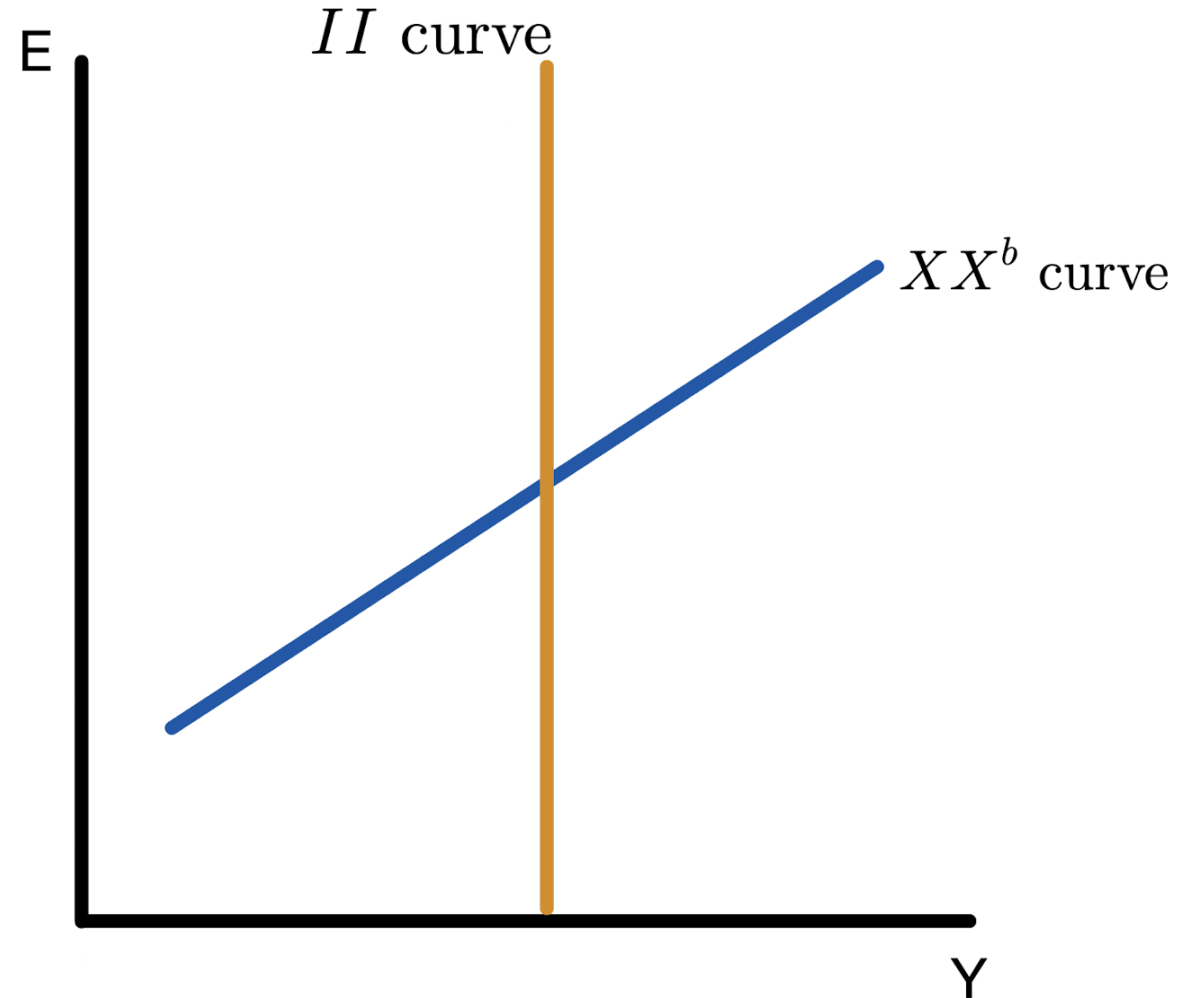
# Internal and External Balance

- *II* curve:  $(Y, E)$  consistent with internal balance  $Y = Y^f$

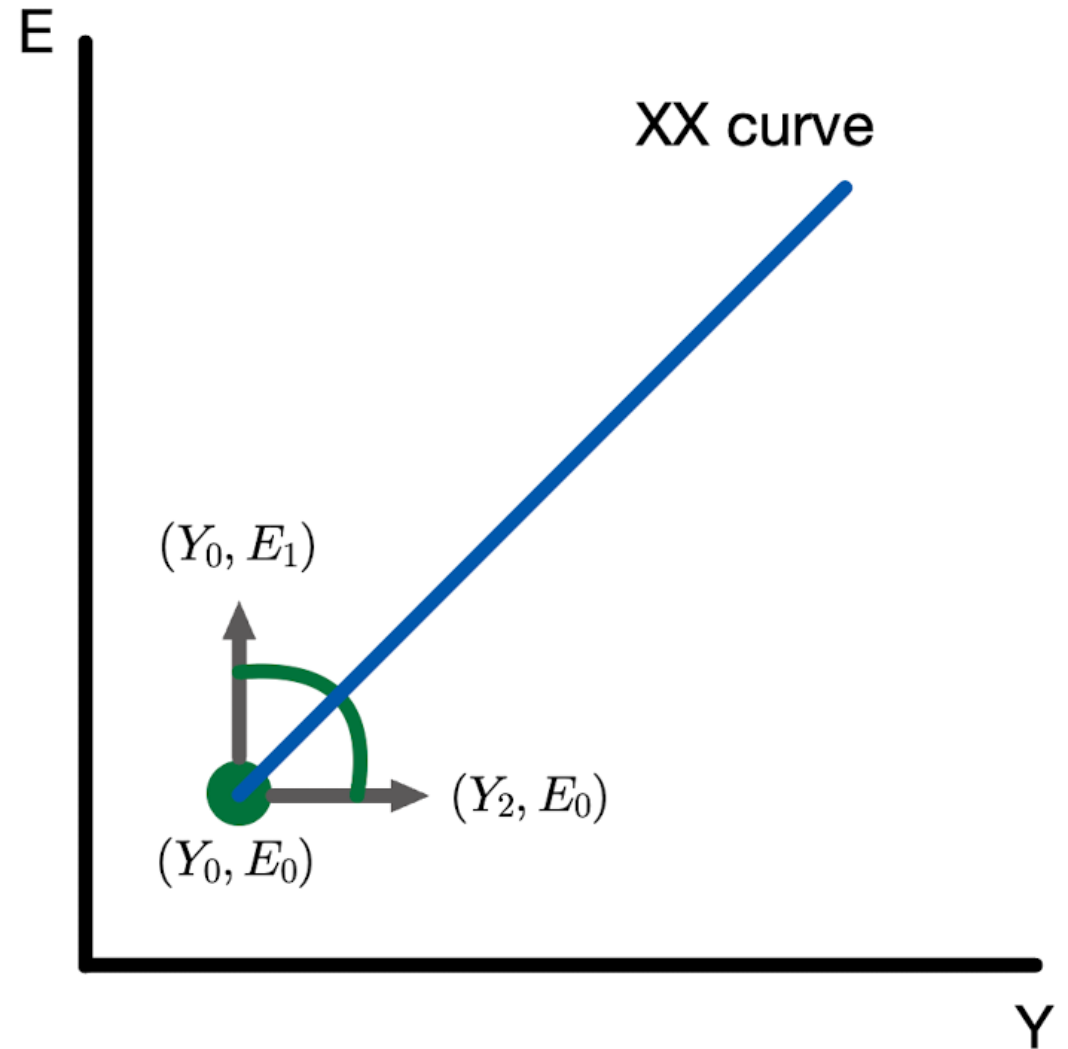


# Internal and External Balance

- *II* curve:  $(Y, E)$  consistent with **internal balance**  $Y = Y^f$
- *XX* curve:  $(Y, E)$  consistent with **external balance**  $CA = XX^b$

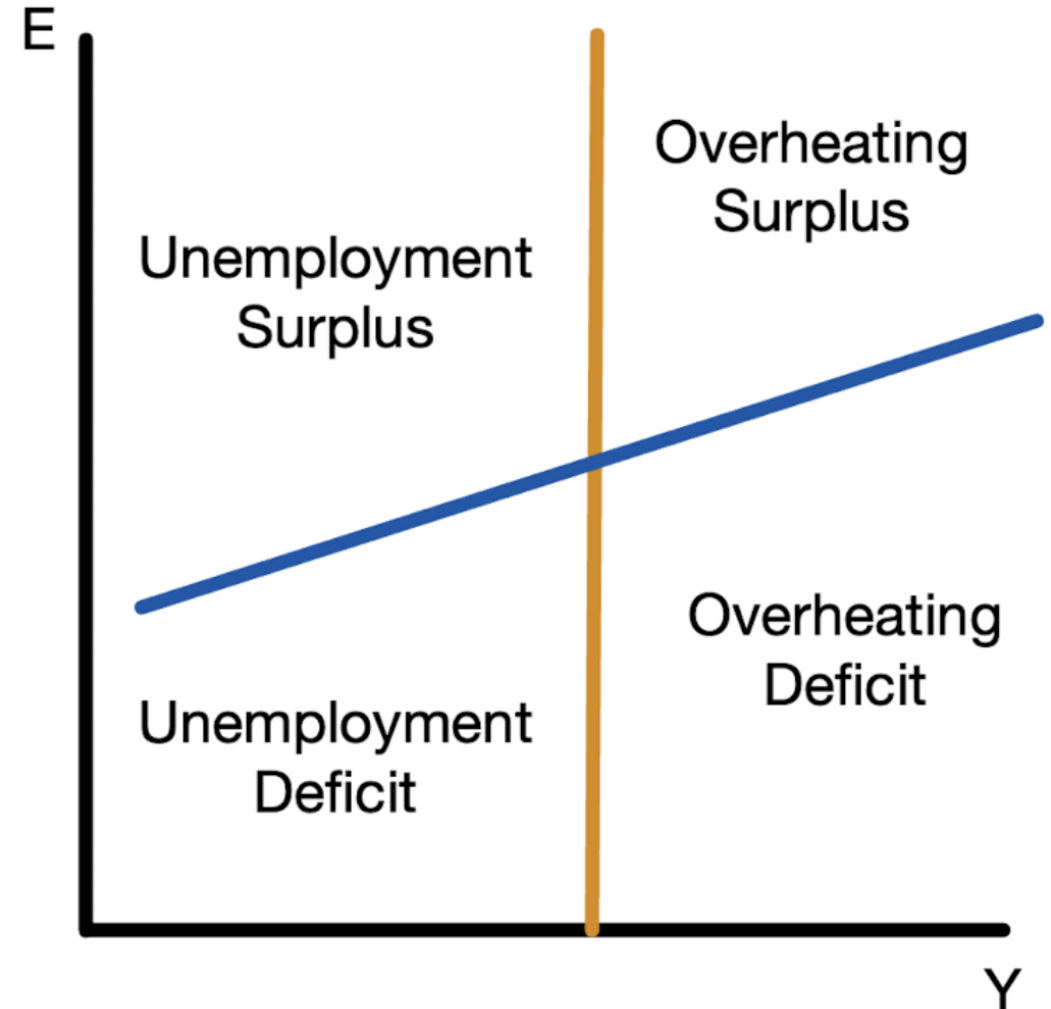


# The XX curve is upward sloping



# Four Zones of Discomfort

- Lack of internal balance:
  - Overheating when  $Y > Y^f$
  - Unemployment when  $Y < Y^f$
- Lack of external balance:
  - Deficit when  $CA < XX^b$
  - Surplus when  $CA > XX^b$
- Good tool to diagnose state of an economy

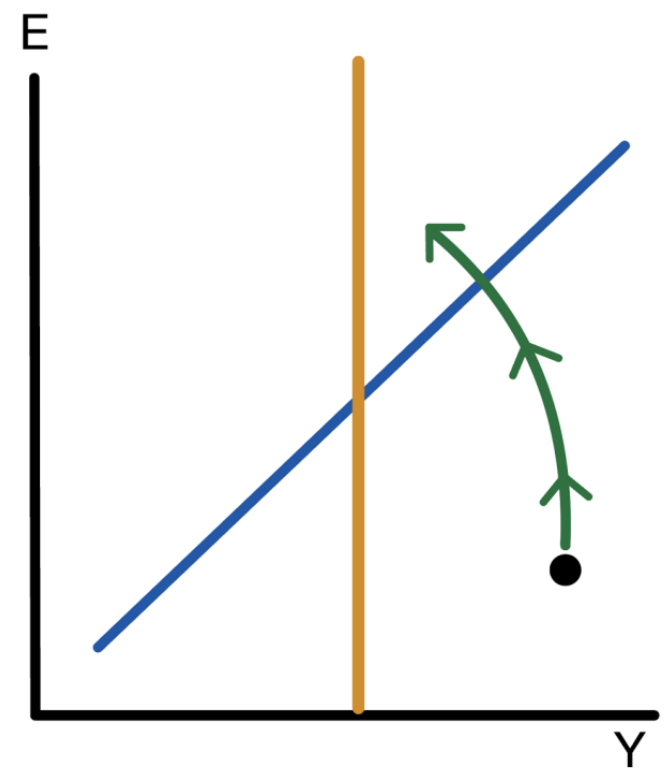
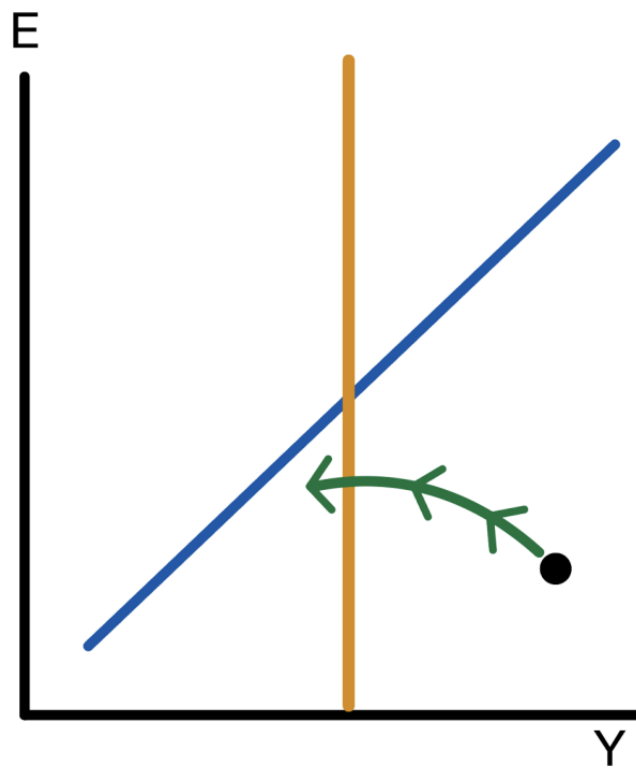
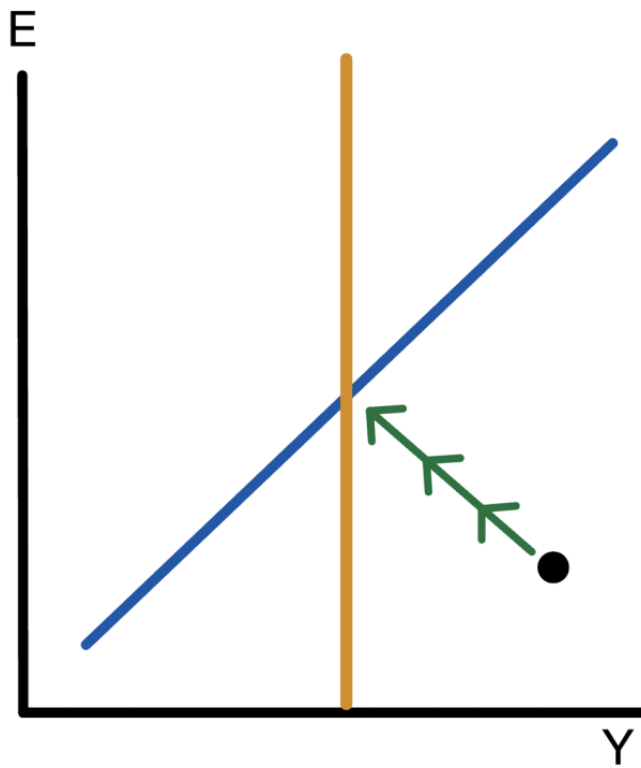


# Dynamics

- Output adjusts to  $Y^f$  in the long run
- Current account adjusts to  $XX^b$  in the long run
- Otherwise, dynamics unspecified
- Can take a long time to return to balance
- Government policy can speed up return to balance

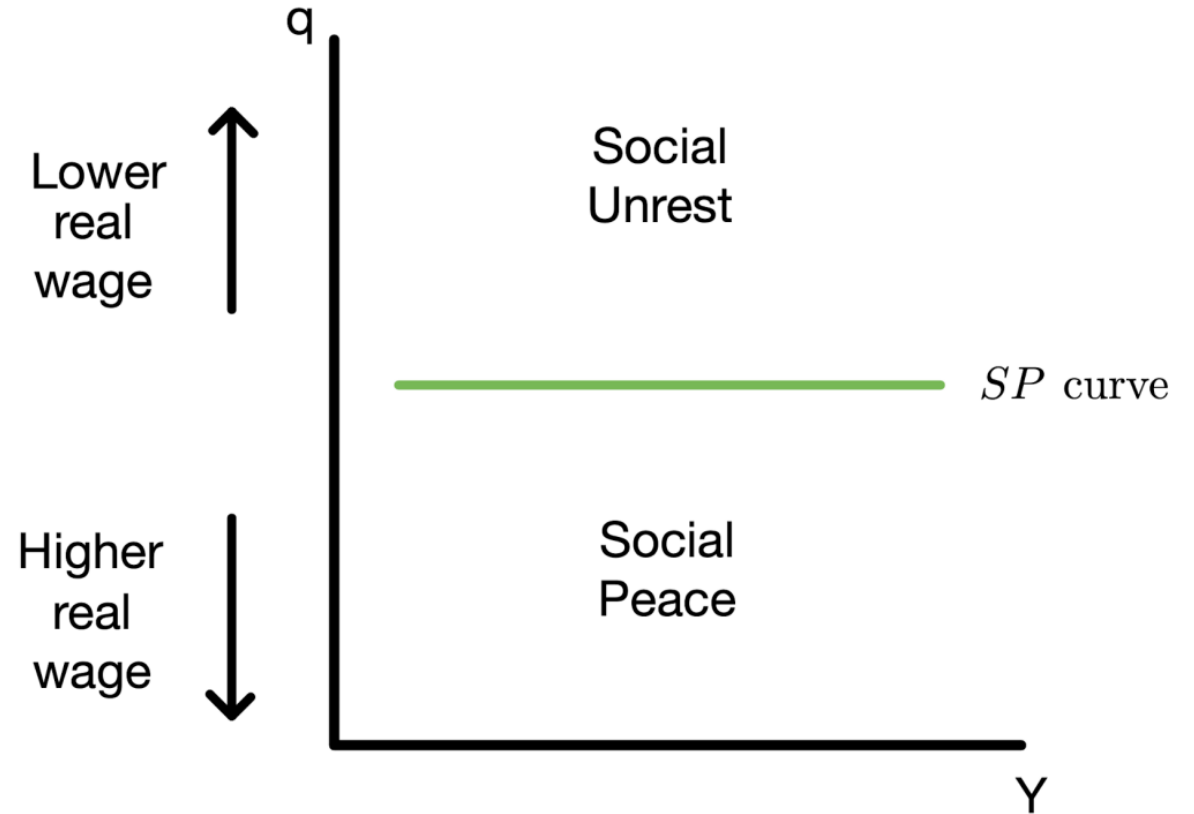


# Possible Dynamics



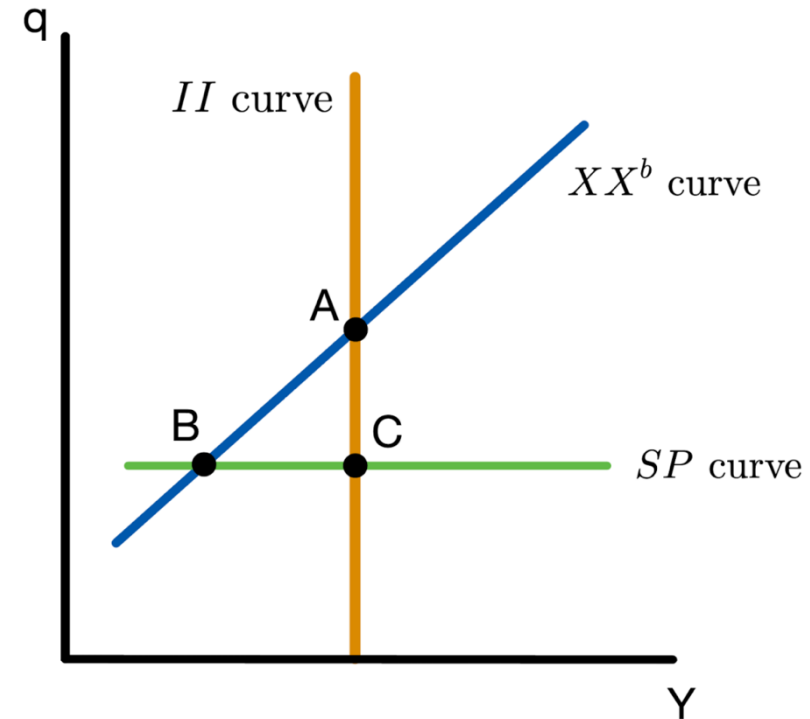
# Social Peace

- When real wages are too low, there is **social unrest**
- Real depreciation associated with lower real wage



# Economic and Social Objectives

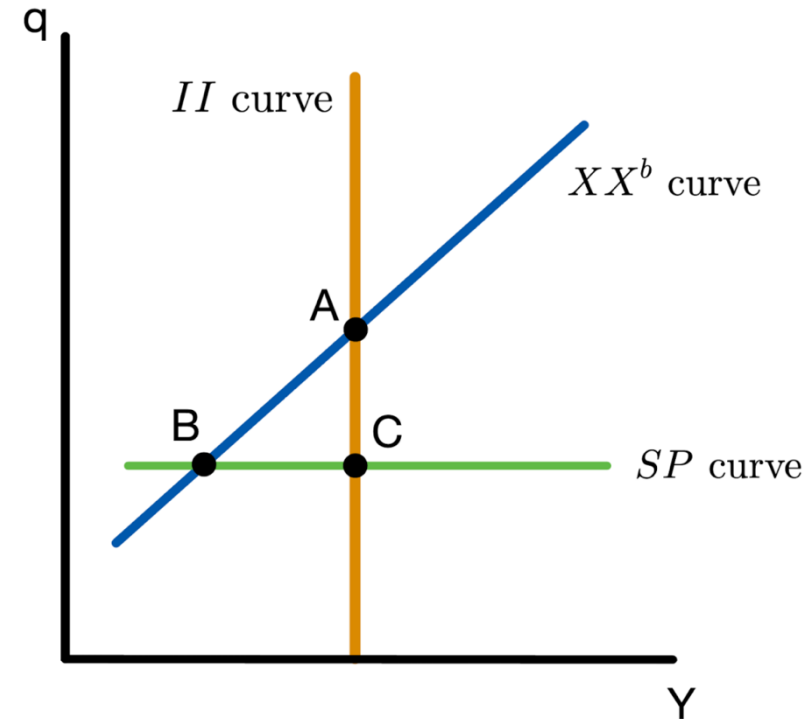
- When SP curve is low enough, social and economic objectives are in conflict
- Assign evocative labels to A, B, C.



| Point | Internal Balance | External Balance | Social Peace | Name |
|-------|------------------|------------------|--------------|------|
| A     | ✓                | ✓                | Conflict     |      |
| B     | Unemployment     | ✓                | ✓            |      |
| C     | ✓                | Deficit          | ✓            |      |

# Economic and Social Objectives

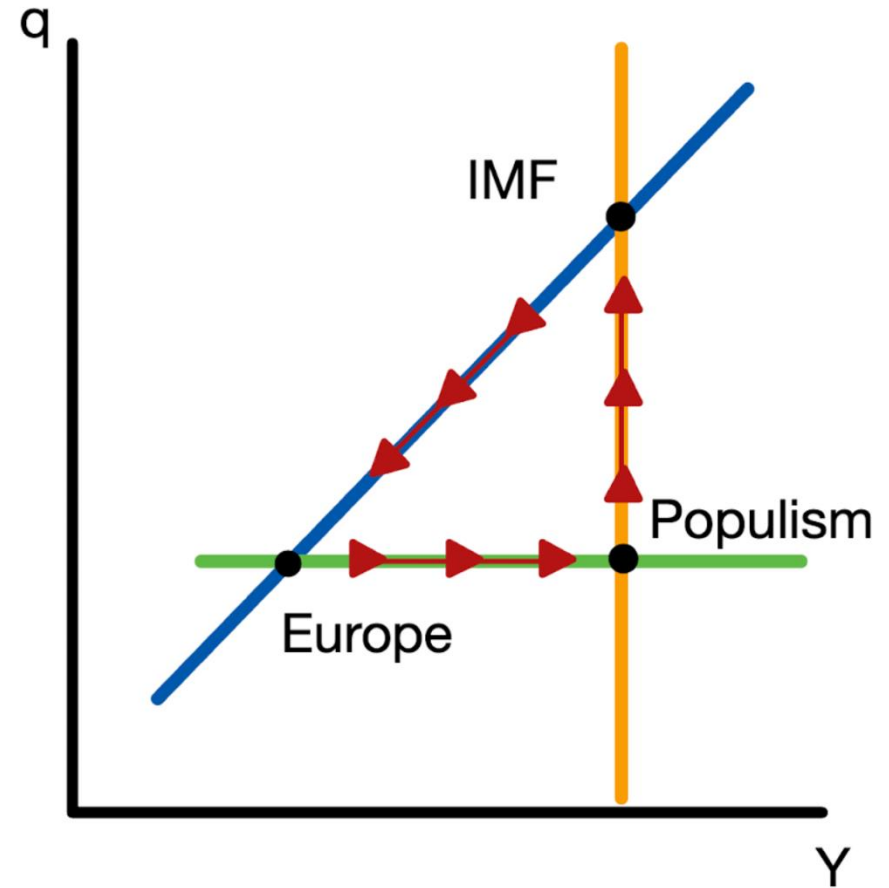
- When SP curve is low enough, social and economic objectives are in conflict
- Assign evocative labels to A, B, C.



| Point | Internal Balance | External Balance | Social Peace | Name     |
|-------|------------------|------------------|--------------|----------|
| A     | ✓                | ✓                | Conflict     | IMF      |
| B     | Unemployment     | ✓                | ✓            | Europe   |
| C     | ✓                | Deficit          | ✓            | Populism |

# The Latin Triangle

- Economy cycles between IMF, Europe, Populism
- Describes dynamics in Latin America *scarily* well
- Economic cycles and how society sees itself
- No easy solutions



# The $II$ - $XX$ with Domestic Demand and Real Exchange Rates

- Assume specific adjustment mechanisms to pin down dynamics
- Adjust to internal balance through the price level  $P$
- Adjust to external balance through consumption  $C$
- Study environmental sustainability (together with social peace and economic balance)

# The II Curve

- Domestic demand

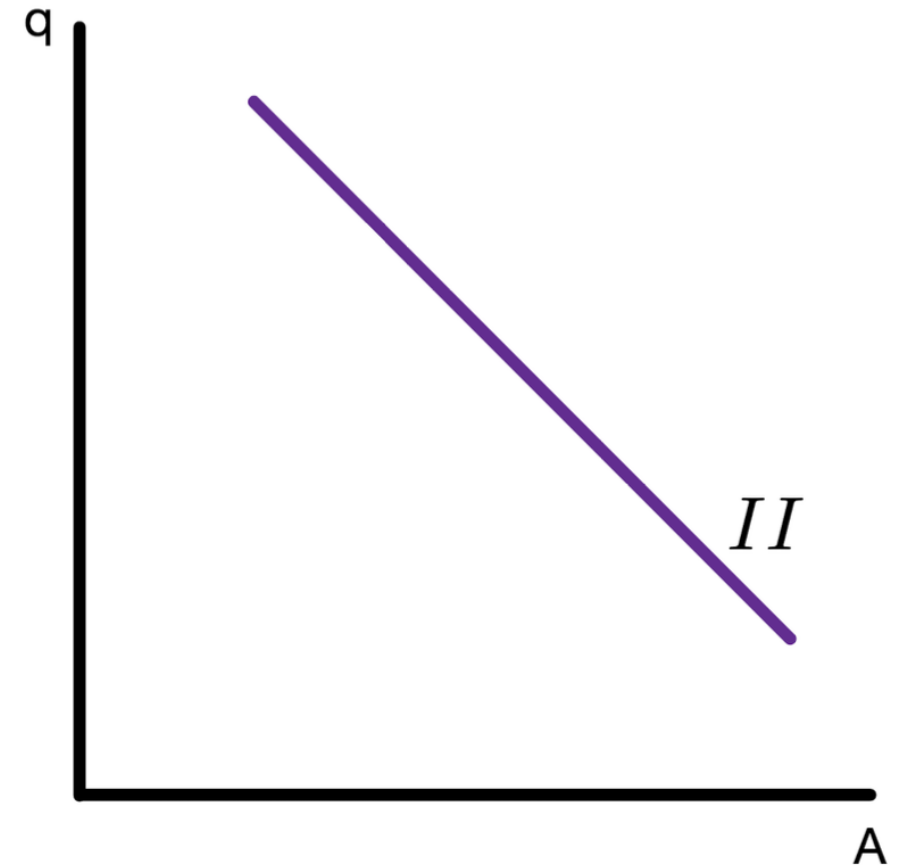
$$A = C + I + G$$

$$= C(Y - T) + I + G$$

- Demand for domestic goods

$$Z = C + I + G + CA$$

$$= C(Y - T) + I + G + CA(q, Y, Y^*)$$



# The II Curve

- Domestic demand

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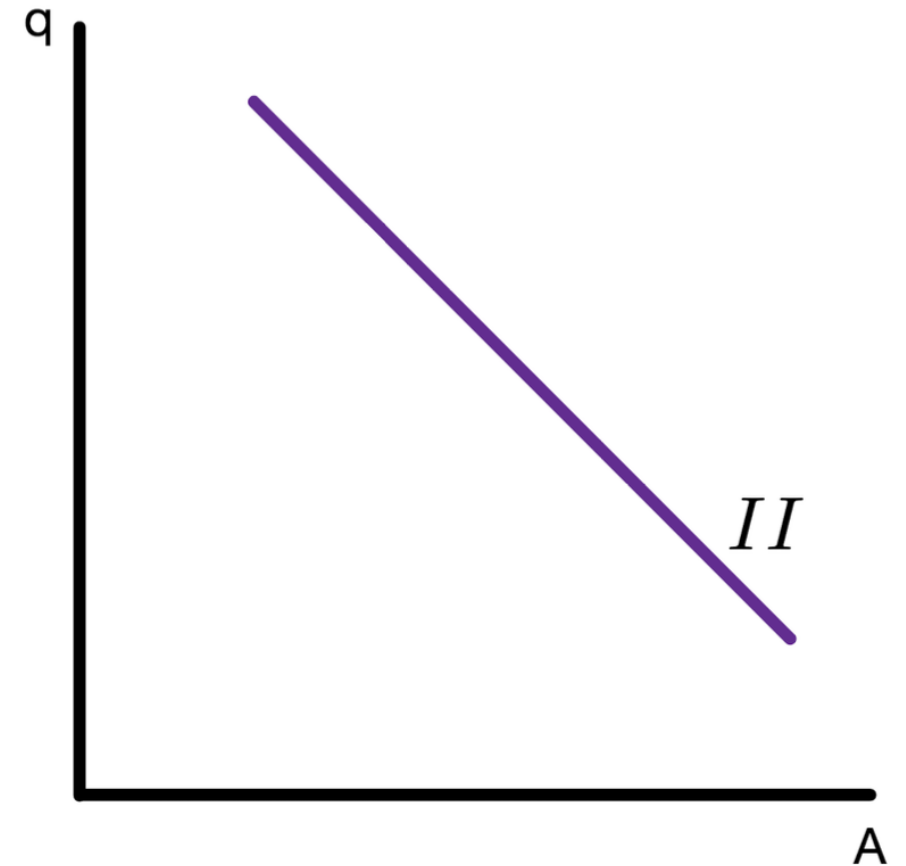
- Demand for domestic goods

$$Z = C + I + G + CA$$

$$= C(Y - T) + I + G + CA(q, Y, Y^*)$$

- Combine both

$$A = Y - CA(q, Y, Y^*)$$





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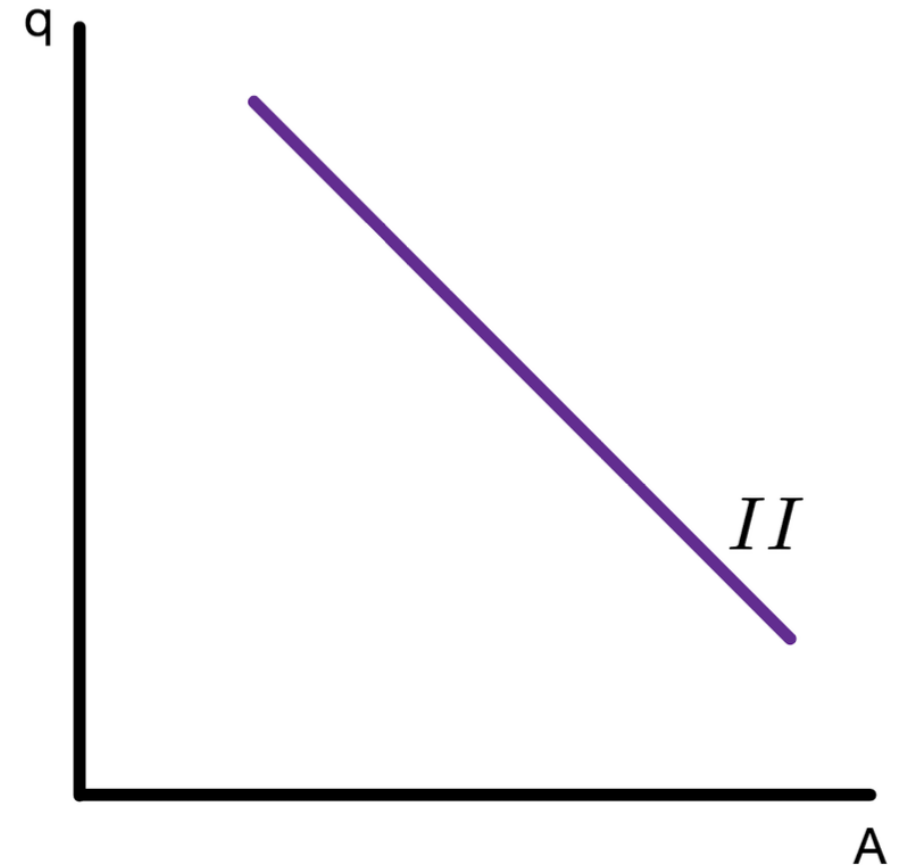
$$\begin{aligned} Z &= C + I + G + CA \\ &= C(Y - T) + I + G + CA(q, Y, Y^*) \end{aligned}$$

- Combine both

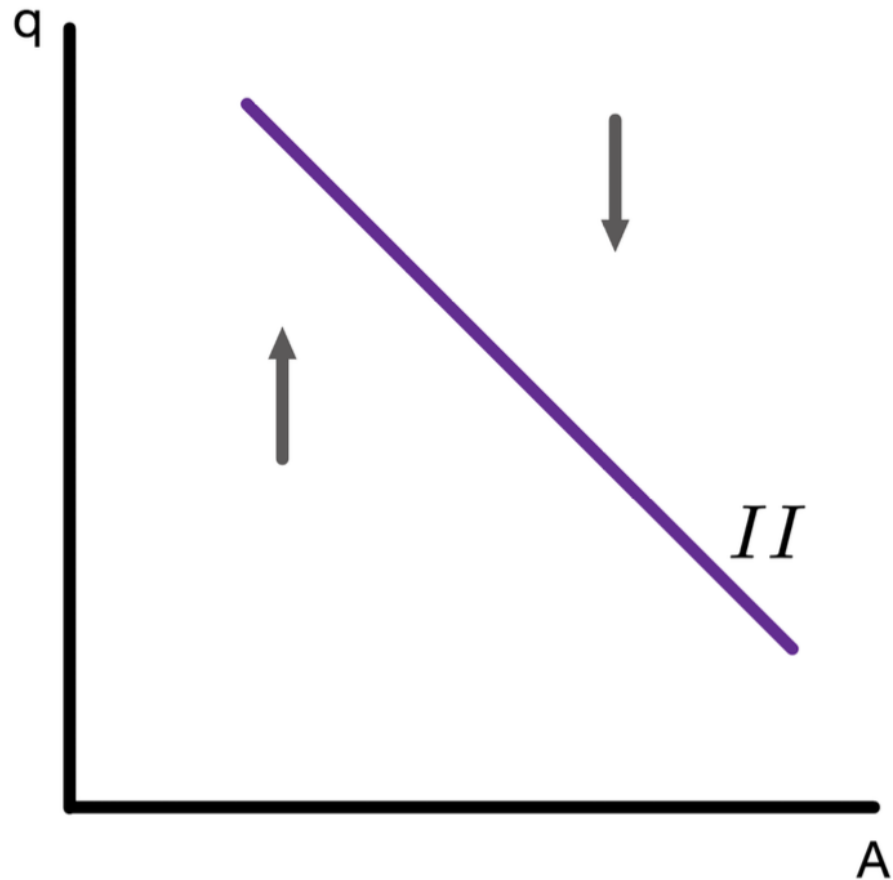
$$A = Y - CA(q, Y, Y^*)$$

- Set  $Y = Y^f$

$$A = Y^f - CA(q, Y^f, Y^*)$$



# Adjustment Dynamics for II Curve

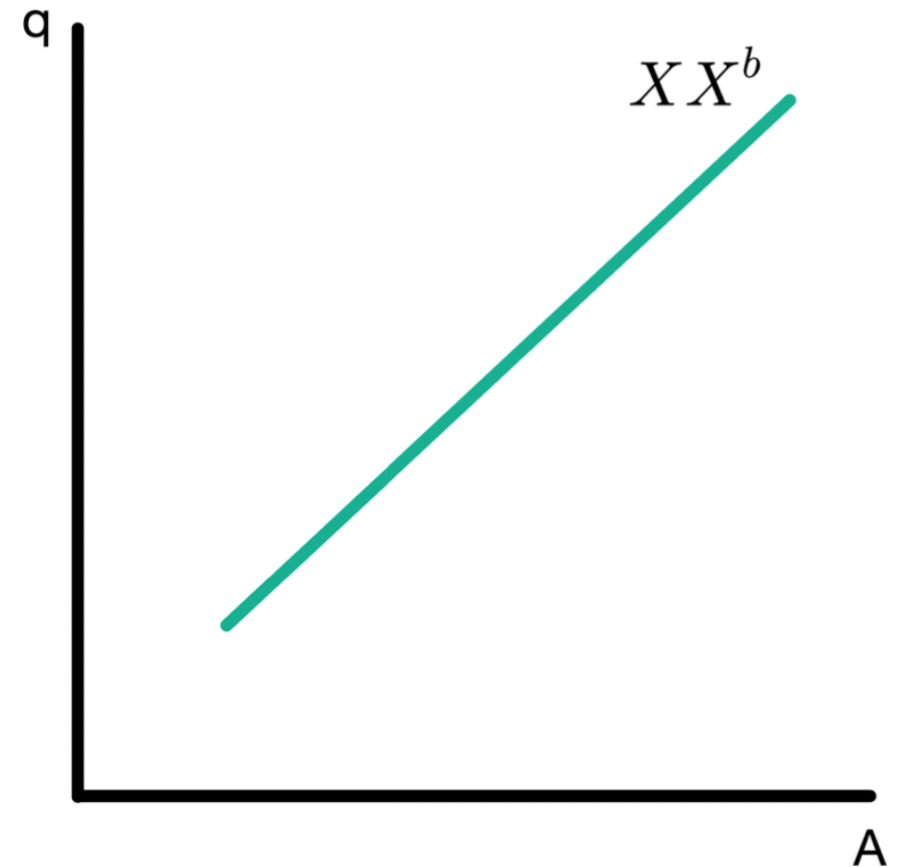


- The economy tends toward internal balance through adjustments in the price level  $P$ 
  - When  $Y > Y^f$ ,  $P$  is increasing
  - When  $Y < Y^f$ ,  $P$  is decreasing
- Changes in  $P$  make  $Y$  approach  $Y^f$

# The $XX$ Curve

Start with domestic demand

$$A = C(Y - T) + I + G$$



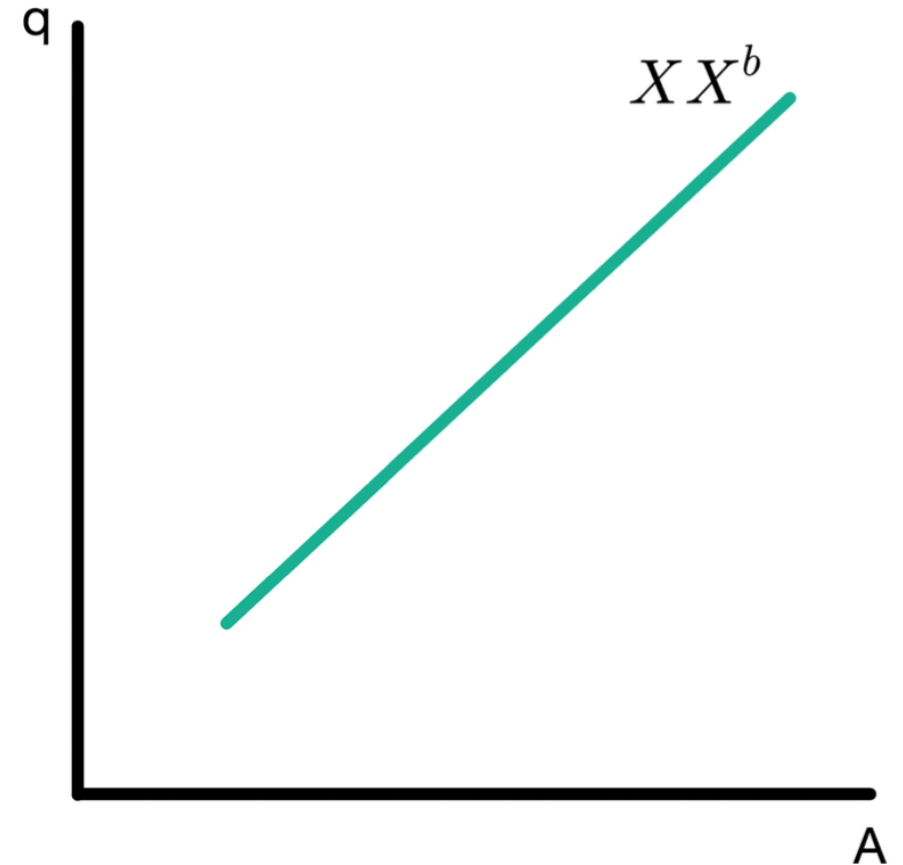
# The XX Curve

Start with domestic demand

$$A = C(Y - T) + I + G$$

Solve for Y

$$Y = C^{-1}(A - I - G) + T$$



# The XX Curve

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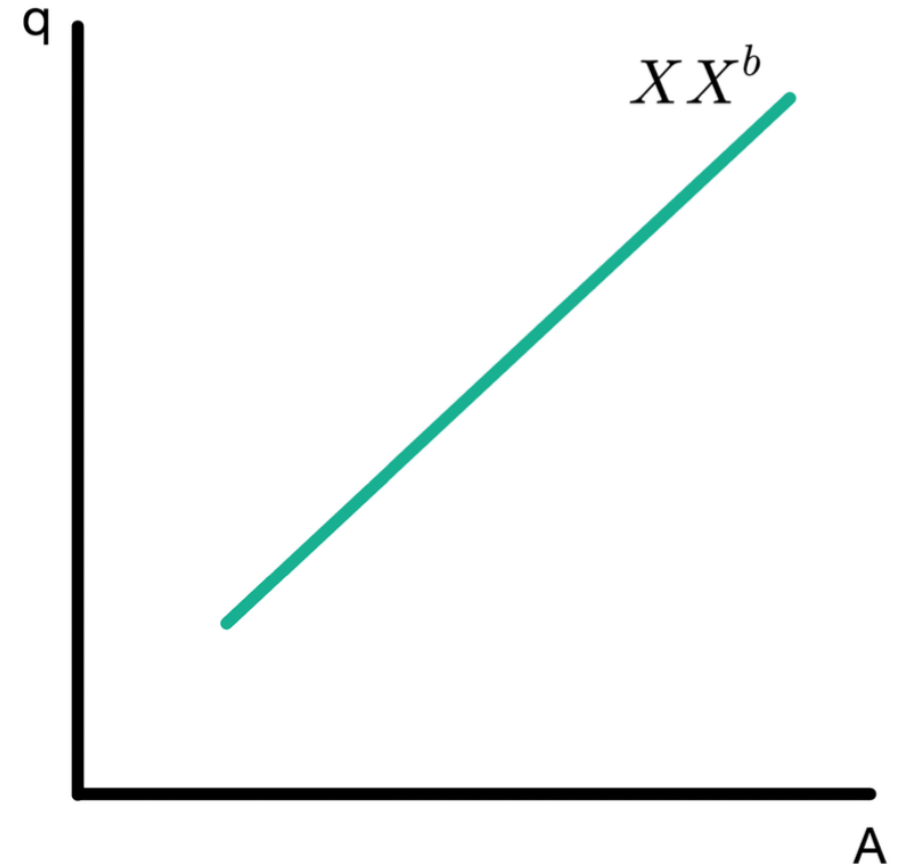
$$A = C(Y - T) + I + G$$

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Insert in

$$\begin{aligned} XX^b &= CA(q, Y, Y^*) \\ &= CA(q, C^{-1}(A - I - G) + T, Y^*) \end{aligned}$$



# The XX Curve

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Solve for Y

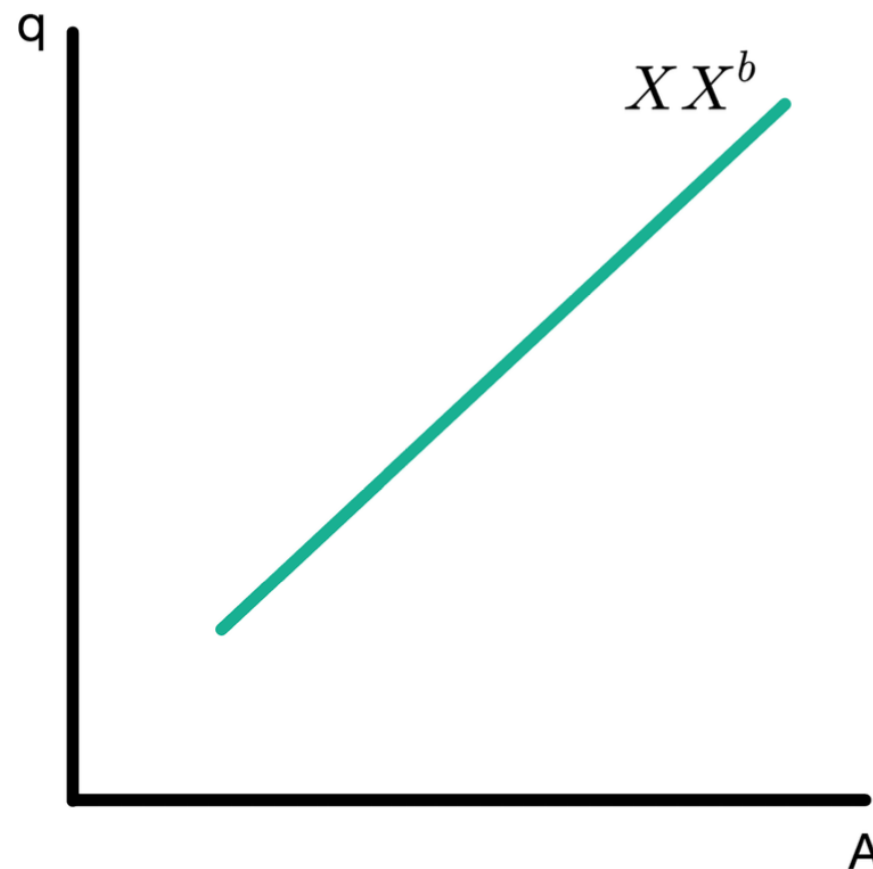
$$Y = C^{-1}(A - I - G) + T$$

Insert in

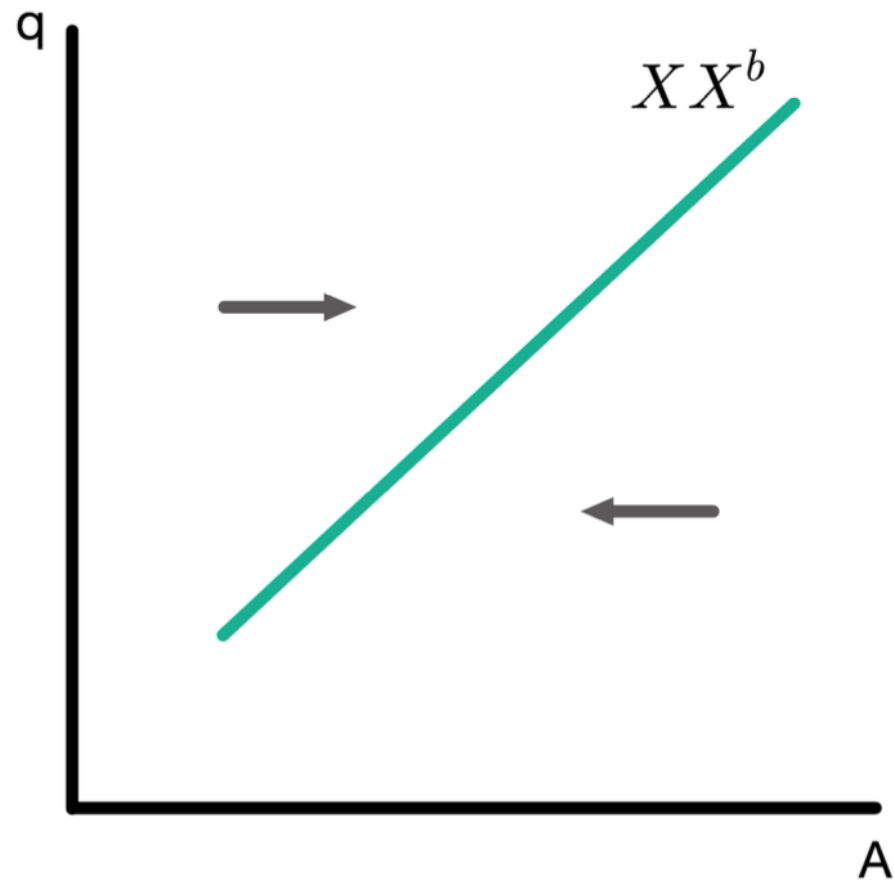
$$\begin{aligned} XX^b &= CA(q, Y, Y^*) \\ &= CA(q, C^{-1}(A - I - G) + T, Y^*) \\ &= \widetilde{CA}(q, A, Y^*) \end{aligned}$$

where the function  $\widetilde{CA}$  is defined by

$$\widetilde{CA}(q, A, Y^*) \equiv CA(q, C^{-1}(A - I - G) + T, Y^*)$$

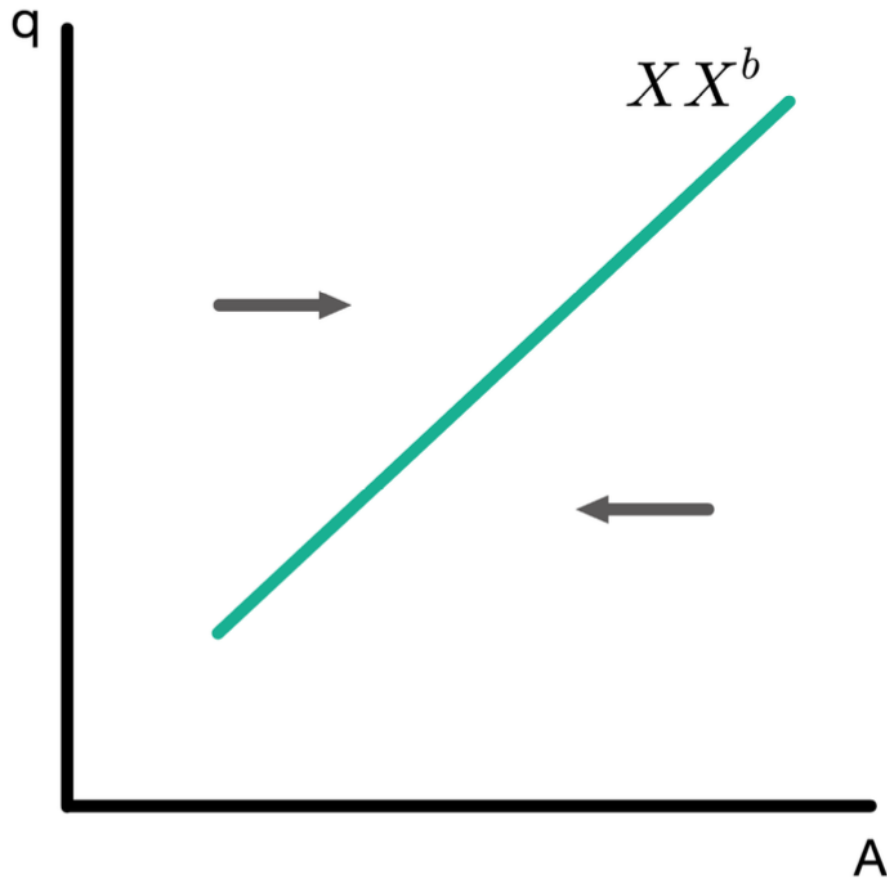


# Adjustment Dynamics for $XX$ Curve



Two sources of adjustment:

# Adjustment Dynamics for $XX$ Curve

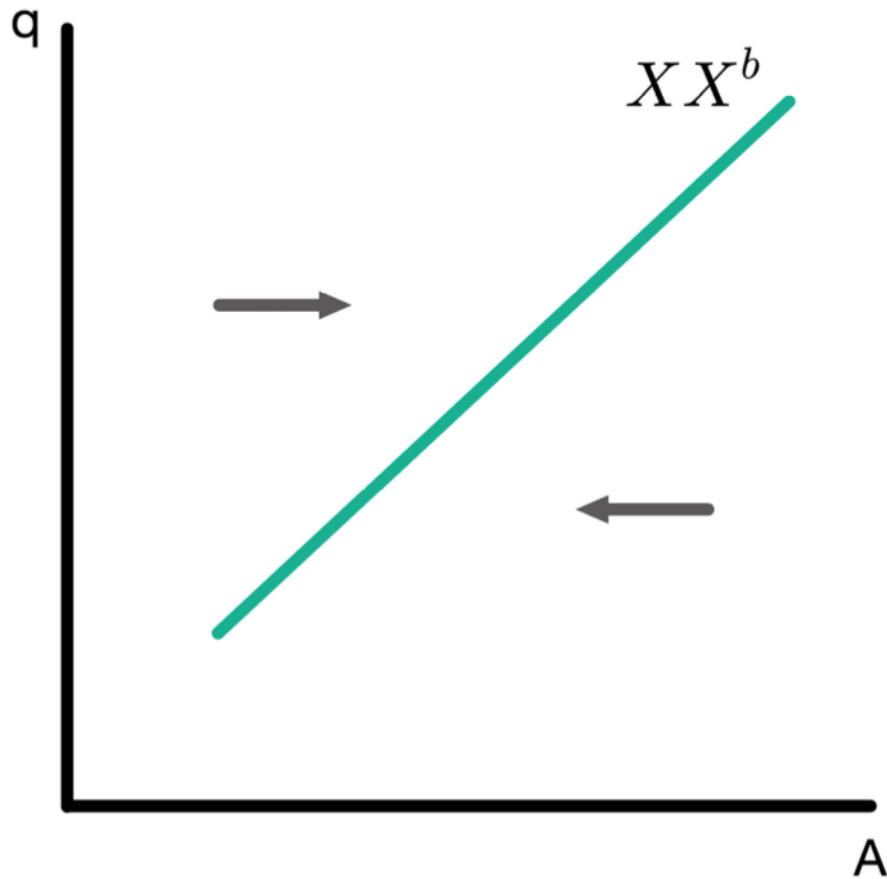


Two sources of adjustment:

- An aggregate budget constraint: we cannot finance too high a level of consumption with ever-increasing borrowing.



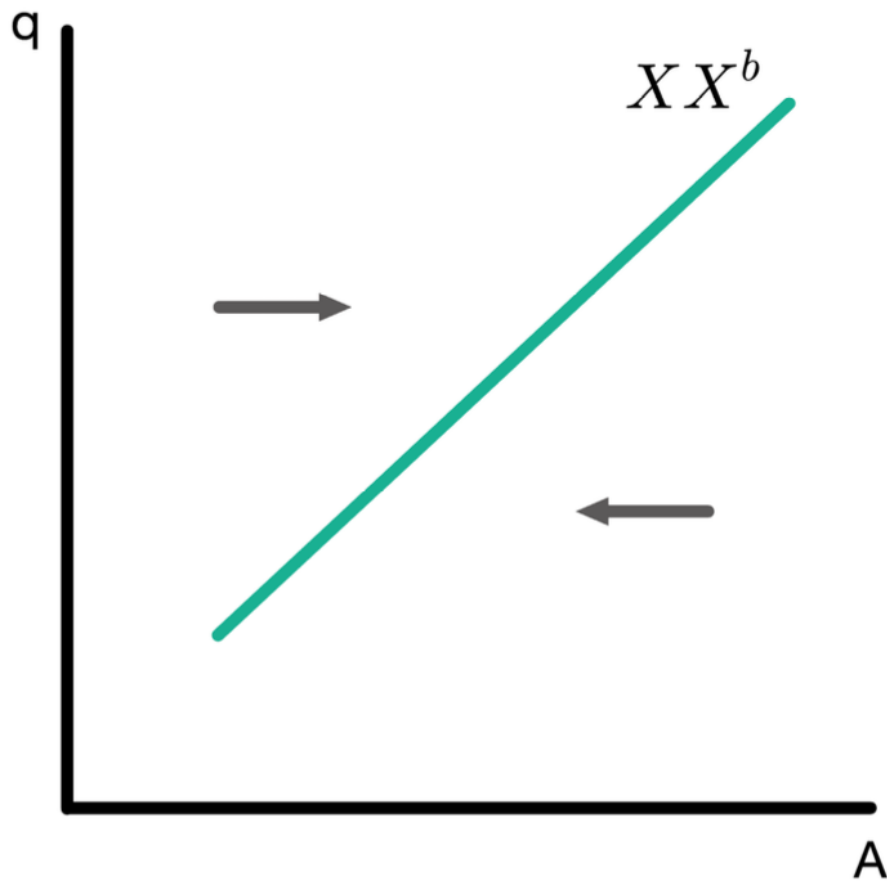
# Adjustment Dynamics for $XX$ Curve



Two sources of adjustment:

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  - When  $CA < XX^b$ , consumption is decreasing over time through budget constraint provide

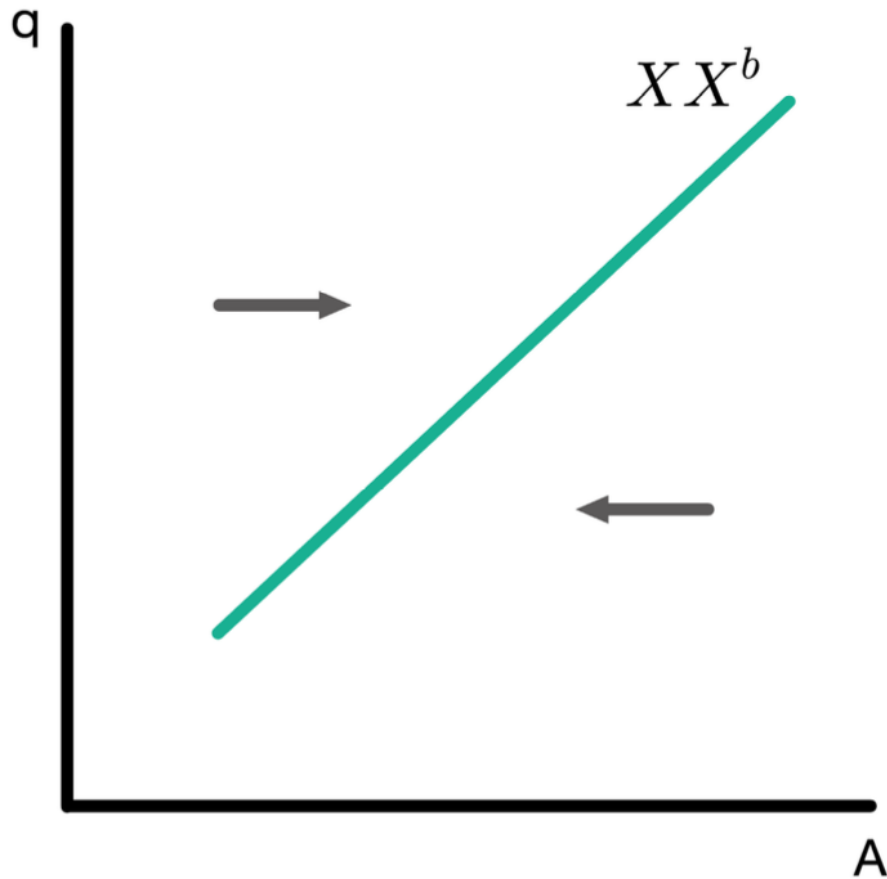
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- We value consumption and will not choose to consume less in every period if it can consume more (nonsatiation)

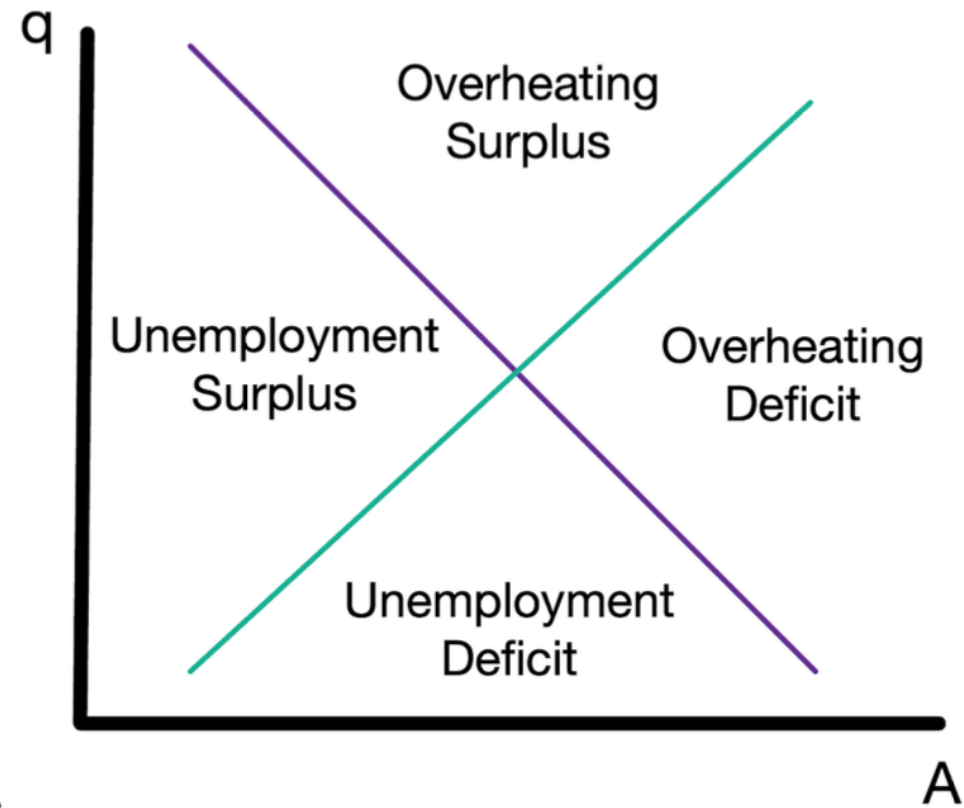
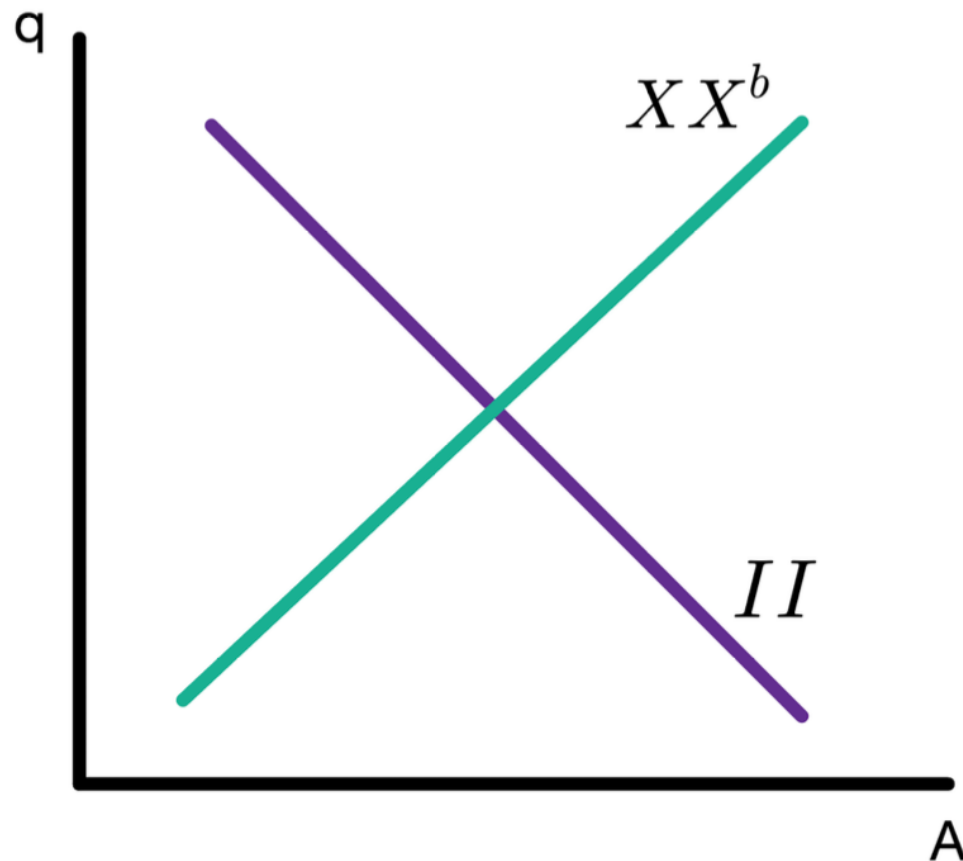
# Adjustment Dynamics for $XX$ Curve



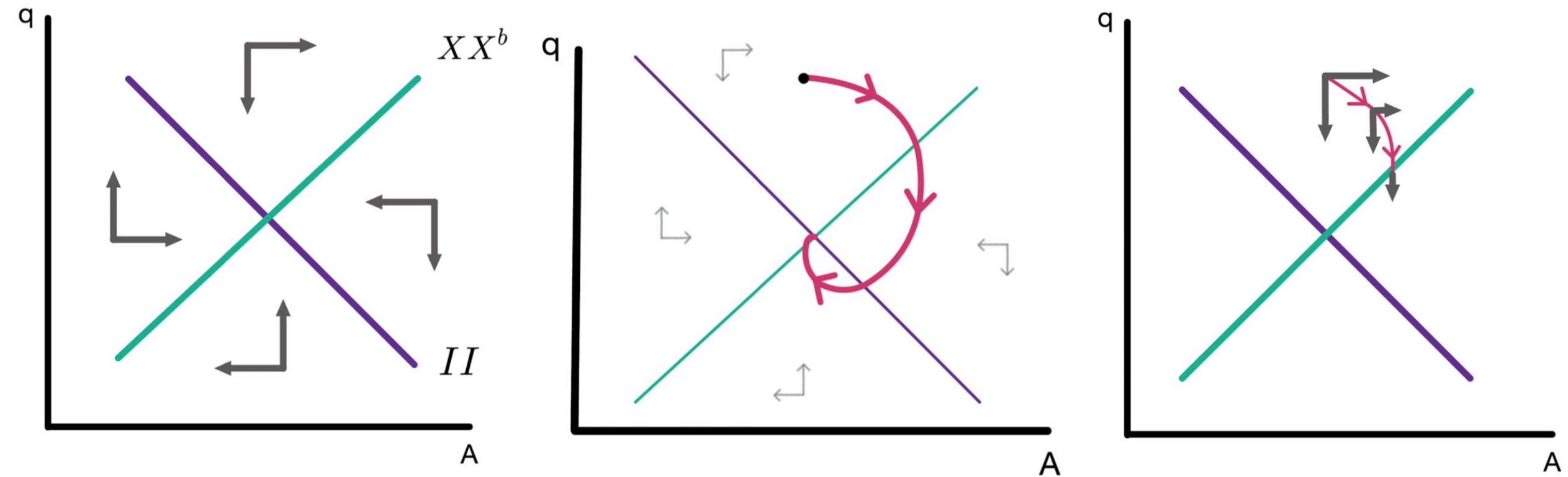
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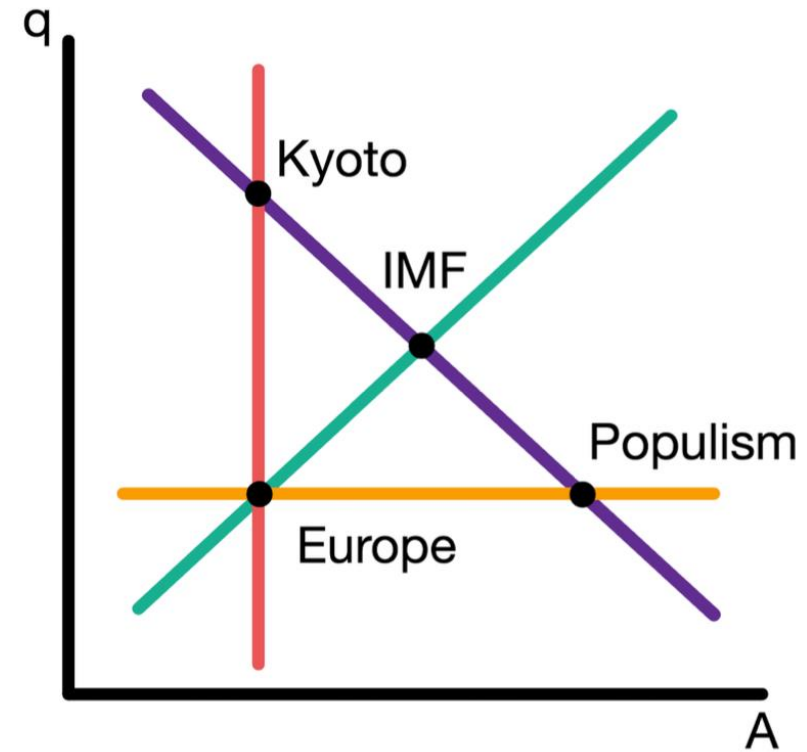
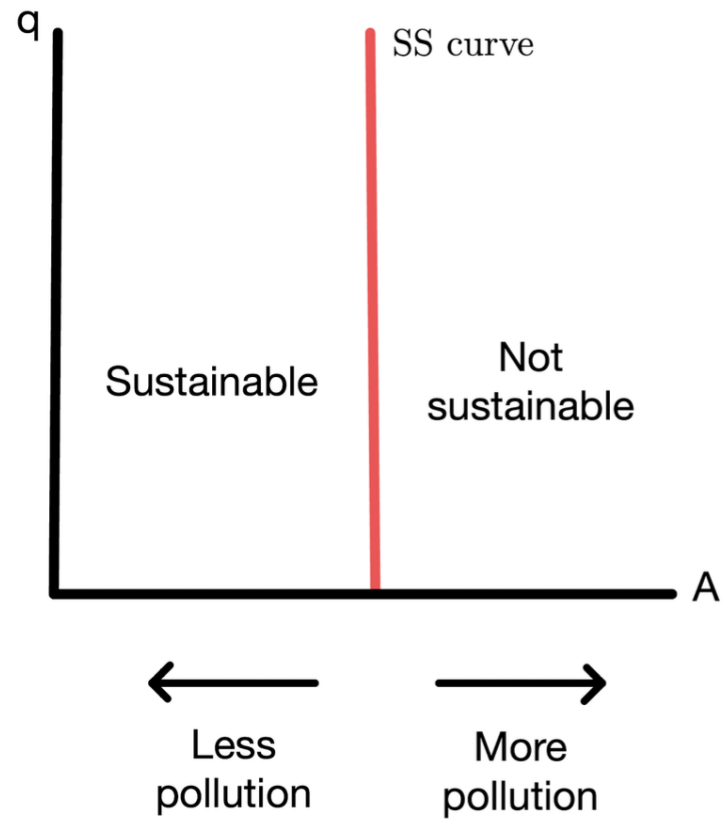
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- We value consumption and will not choose to consume less in every period if it can consume more (nonsatiation)
  - When  $CA > XX^b$ , consumption is increasing over time through nonsatiation

# Four Zones of Economic Discomfort



# Dynamics for II and $XX^b$ Combined

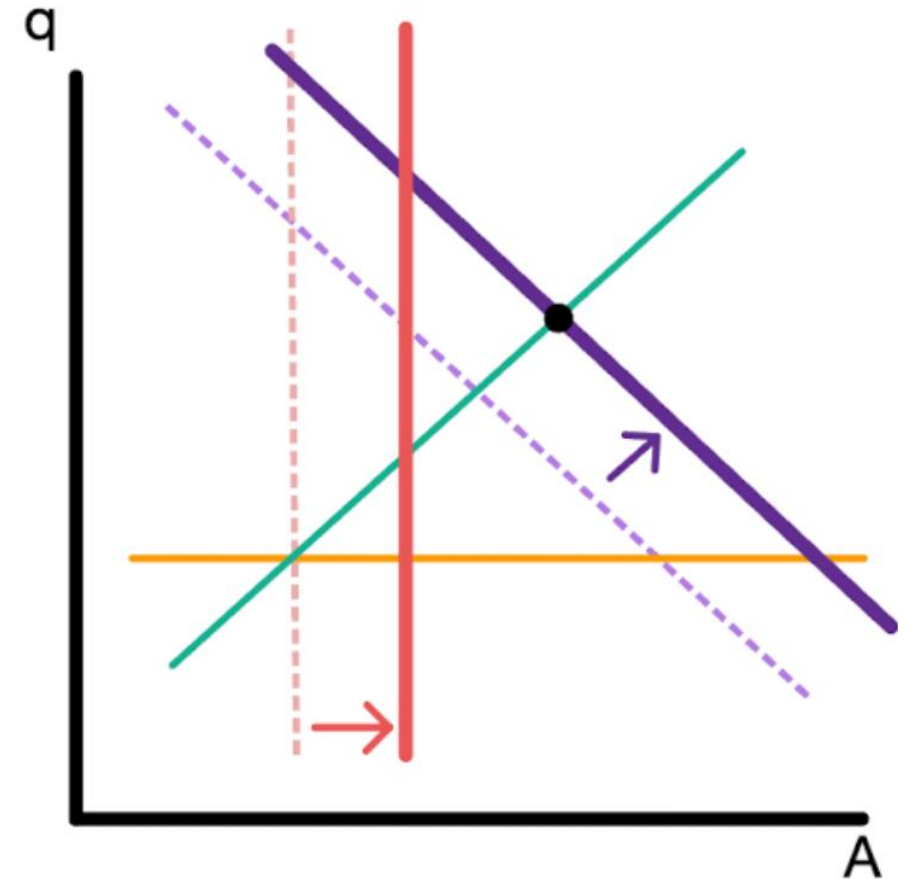




|          | Internal Balance | External Balance | Social Peace | Environment    |
|----------|------------------|------------------|--------------|----------------|
| IMF      | ✓                | ✓                | Conflict     | Pollution      |
| Europe   | Unemployment     | ✓                | ✓            | ✓              |
| Populism | ✓                | Deficit          | ✓            | High Pollution |
| Kyoto    | ✓                | Surplus          | Conflict     | ✓              |

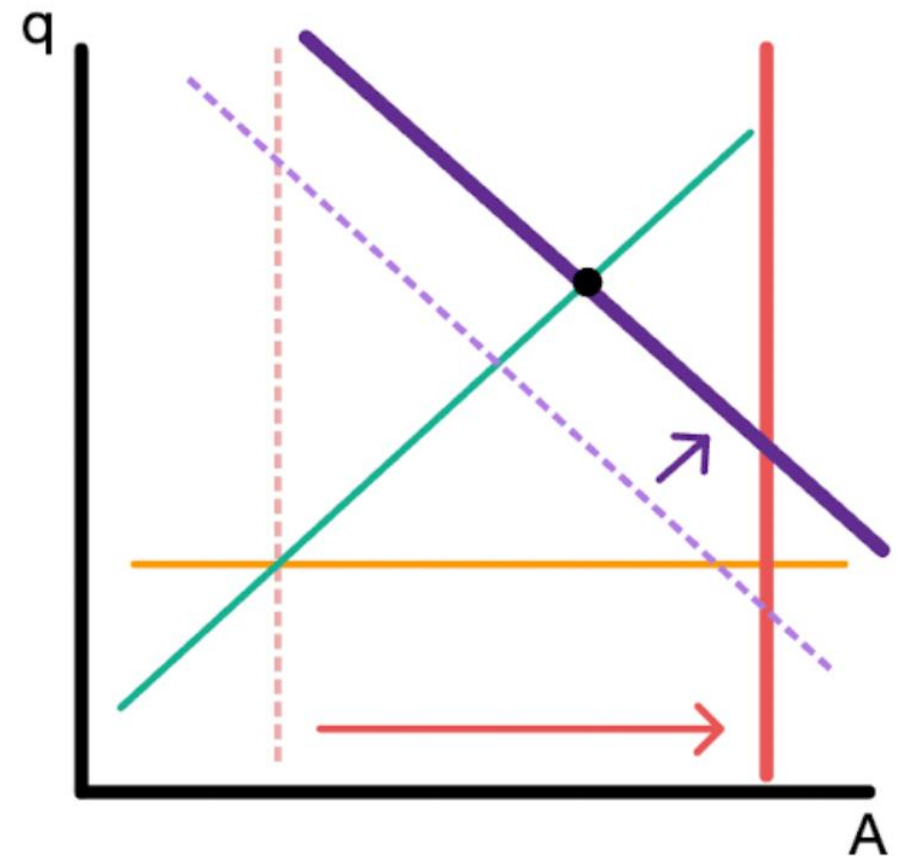
# Can Technological Progress Help?

- It depends!
- Technological progress:  $A$  in the production function  $Y = AK^{1-\alpha} N^\alpha$ 
  - Can help us produce more without harming the environment (SS shifts to the right)
  - But also makes  $Y^f$  go up (II shifts to the right)
- If II shifts a lot and SS shifts little, then environmental sustainability still incompatible with internal and external balance



# Can Technological Progress Help?

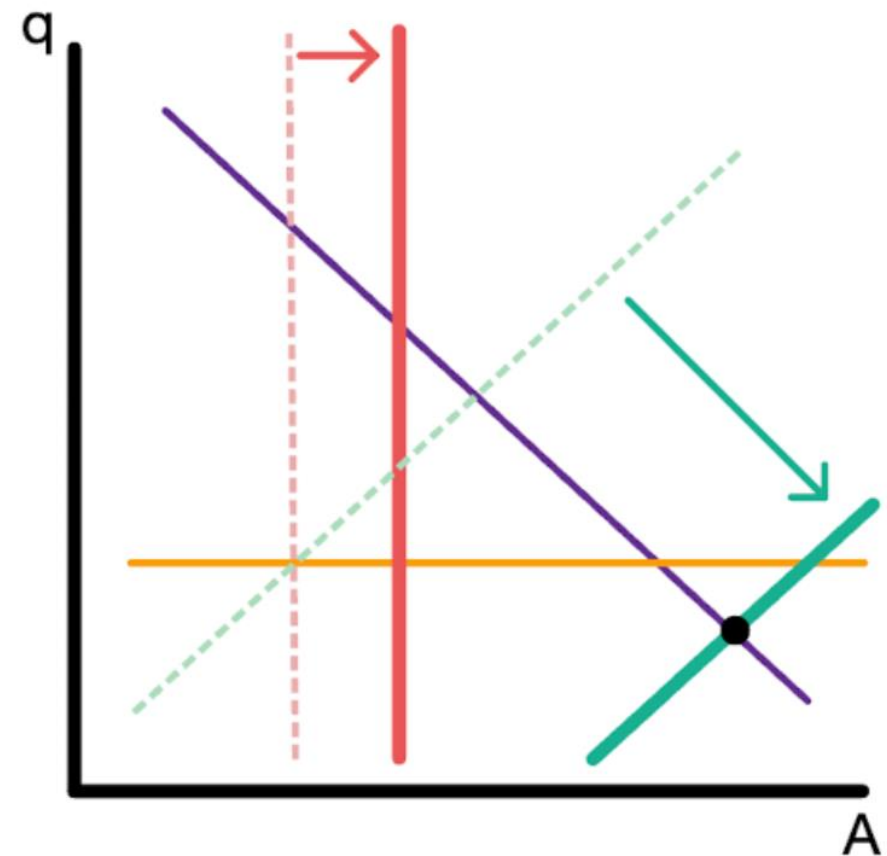
- If  $II$  shifts a little compared to  $SS$ , then environmental sustainability can be achieved together with internal and external balance!





# Can Technological Progress Help?

- Technology can also shift the  $XX$ 
  - For example, higher productivity abroad can increase  $Y^*$
- Technology that make a larger  $CA$  deficit sustainable shift the  $XX$  down
- Large  $XX$  shifts and small  $SS$  shifts do not help solve the conflict between environmental and economic objectives



# Can Technological Progress Help?

- Shifts in  $XX$  that are relative small compared to shifts in  $SS$  can solve the conflict between environmental and economic objectives!

